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Everything About Z-Tests and T-Tests Explained With Pizza and Regret

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Z-Test vs T-Test Explained for Beginners

Every time Netflix changes a thumbnail on your homepage, a statistical test decided it. Not a designer's gut feeling. Not a manager's opinion in a meeting. A test. They showed version A to a few hundred thousand users and version B to another few hundred thousand, counted who clicked, and let the math settle the argument.

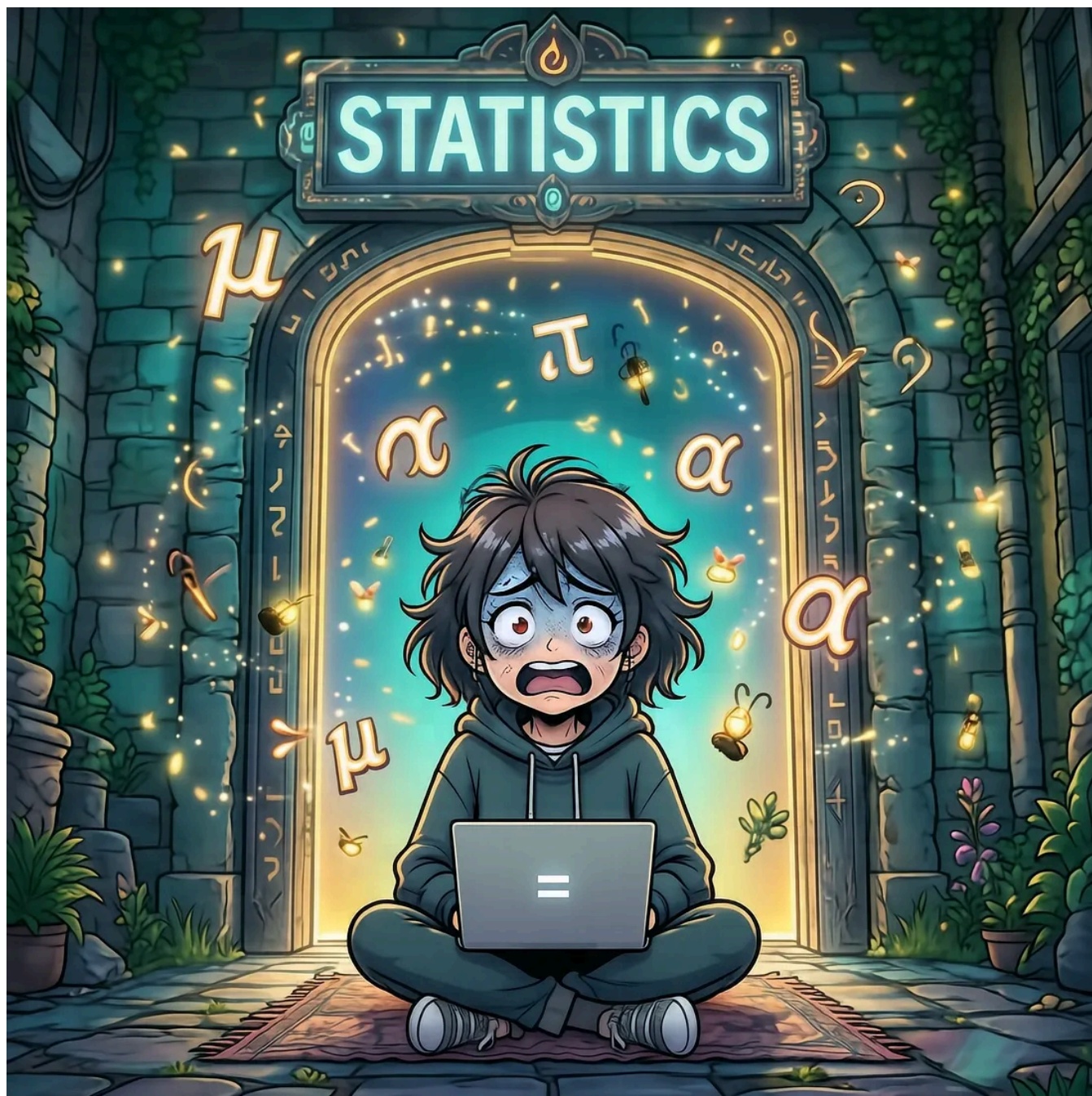
Every drug in your medicine cabinet passed the same kind of test before the FDA said "go ahead, sell it." Every time Instagram tweaks the order of your feed, same thing. Every time a teacher wonders if a new teaching method actually worked, same thing.

You are neck-deep in statistical inference. You just don't have the vocabulary yet.

That changes right now. We're going from zero to genuinely understanding z-tests, t-tests, p-values, hypothesis testing, confidence intervals, and every Greek letter they'll throw at you. I'll write real Python code and explain every single line. I'll use actual examples that make sense. And I'll do it without assuming you remember anything from school.

One rule before we start. If a sentence uses a word you don't know, that's my fault, not yours. I'll explain it immediately or I shouldn't have used it.

Let's go.



You, right now, about to understand something Netflix's data team gets paid six figures to do

The Big Idea (in 30 Seconds)

You want to know the average height of everyone in India. All 1.4 billion people. You could fly there, bring a measuring tape, and spend the next 40 years knocking on

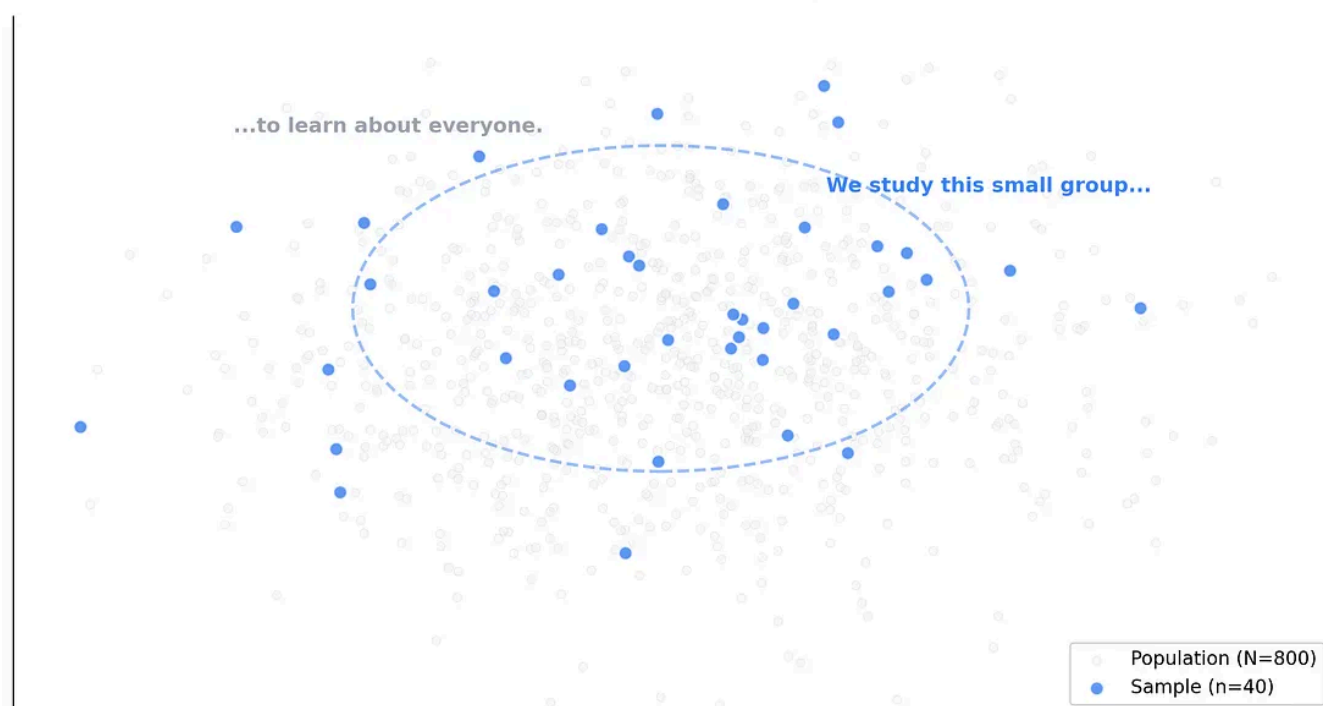
doors. Or you could measure 500 random people and make an educated guess about the other 1,399,999,500.

That's statistical inference. The entire field. You look at a small chunk of something, and you make a careful, math-backed conclusion about the whole thing.

The small chunk is called a **sample**. The whole thing is called a **population**. And the process of jumping from one to the other (with guardrails so you don't land face-first) is inference.

Every single test, formula, and concept in this article exists to answer one question. "Can I trust what my small chunk is telling me about the big picture?" That's the only question. Everything else is machinery.

Population vs Sample



800 dots is the population. 40 highlighted dots is your sample. Your entire job is to figure out if those 40 are trustworthy.

Population, Sample, and the Greek Letters That Scare People

Let's get the vocabulary out of the way because it's simpler than it looks.

Population is every single member of the group you care about. All the customers. All the patients. All the batteries rolling off a factory line. Statisticians use capital N

for the population size. And they use Greek letters for population measurements. The population average is μ (mu, pronounced “myoo”). The population spread is σ (sigma).

Sample is the smaller group you actually measured. Lowercase n for the size. Regular English letters for the measurements. The sample average is $\bar{x}$ (read as “x-bar”). The sample spread is s.

Here’s the cheat sheet you’ll need for the rest of this article.

+-----+	+-----+	+-----+
What	Population	Sample
+-----+	+-----+	+-----+
Average	μ (mu)	x-bar
Spread	σ (sigma)	s
How many	N	n
+-----+	+-----+	+-----+

Greek letters = population. English letters = sample. That’s the entire system. You’ll absorb it naturally if you keep reading.

Standard Deviation. The Most Useful Number Nobody Explains Properly.
Before we touch any test, you need to understand standard deviation. Every formula we’ll see later uses it, and most articles just toss the symbol σ at you and move on. I won’t.

Imagine five friends comparing their monthly rent.

Person A pays \$1,000. B pays \$1,050. C pays \$1,100. D pays \$1,150. E pays \$1,200.

The average rent is \$1,100. Now look at how far each person is from that average.

A is \$100 below. B is \$50 below. C is right on it. D is \$50 above. E is \$100 above.

If you average out all those distances (with a squaring step in between to handle the negative signs, then a square root at the end to undo the squaring), you get the standard deviation. For this group, it’s about \$70.71.

Standard deviation is the average distance from the average. That's the plain-English version. A small standard deviation means everyone is bunched together. A large one means people are all over the place.

Now imagine five different friends with rents of \$200, \$600, \$1,100, \$1,600, and \$2,000. Same average (\$1,100) but wildly different spread. The standard deviation here is about \$640.

Same center. Totally different story. Standard deviation captures the “totally different story” part.

Why this matters for testing. When you run a z-test or t-test, you're asking “is my sample average far enough from the expected value to be suspicious?” But “far enough” depends entirely on how spread out the data is. A sample average that's \$10 away from the population mean is a big deal if σ is \$5. It's nothing if σ is \$500. Standard deviation is the ruler you use to measure “far.”



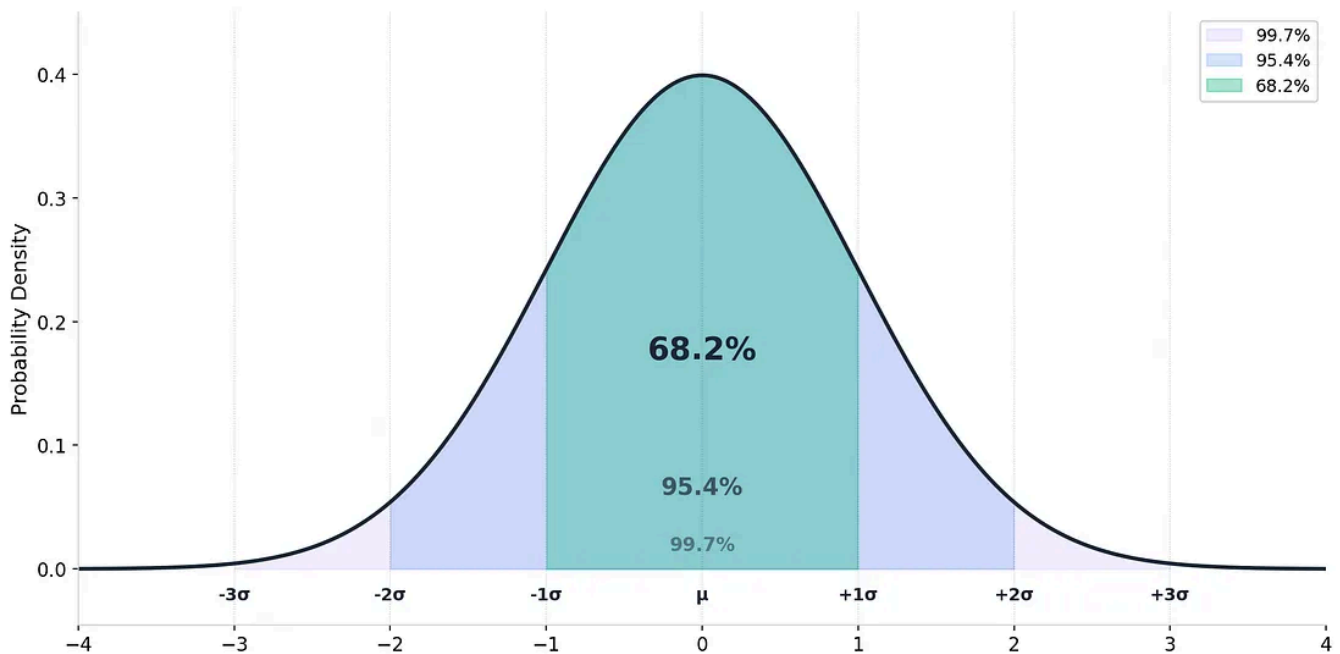
Same average, completely different vibe. That difference is standard deviation.

The Bell Curve. Why It Runs Everything.

Plot the heights of 10,000 random adults on a chart. Most people cluster around the average. Fewer people are a bit taller or a bit shorter. Very few are extremely tall or extremely short. The shape you get is a smooth, symmetric hill.

That shape is the **normal distribution**. You've heard it called the bell curve.

The Normal Distribution (Bell Curve)



68% of everything sits within one standard deviation of the center. That one fact powers half of statistics.

Three properties make this shape special.

It's symmetric. The left side is a mirror of the right. As many people are shorter than average as are taller than average.

The mean sits at the peak. The highest point of the hill is the average value. Most data lives near here.

The 68–95–99.7 rule. This is the killer feature. About 68% of all data falls within 1 standard deviation of the mean. About 95% falls within 2 standard deviations. And 99.7% falls within 3.

Let me make that concrete. If the average adult height is 170 cm and the standard deviation is 10 cm, then roughly 68% of adults are between 160 and 180 cm. About 95% are between 150 and 190 cm. And 99.7% are between 140 and 200 cm. Finding someone outside that 140–200 range? That's 3 out of every 1,000 people. Rare.

This matters because later, when we run a test and get a value that's 3 standard deviations away from the center, we'll know something is off. That kind of distance happens by accident less than 0.3% of the time. So either we got astronomically unlucky, or something real is going on.

That intuition is the entire engine of hypothesis testing. Keep it.

The Central Limit Theorem. The Closest Thing Statistics Has to a Miracle.

This is the single most important idea in the entire article. If you understand this one thing, everything else falls into place.

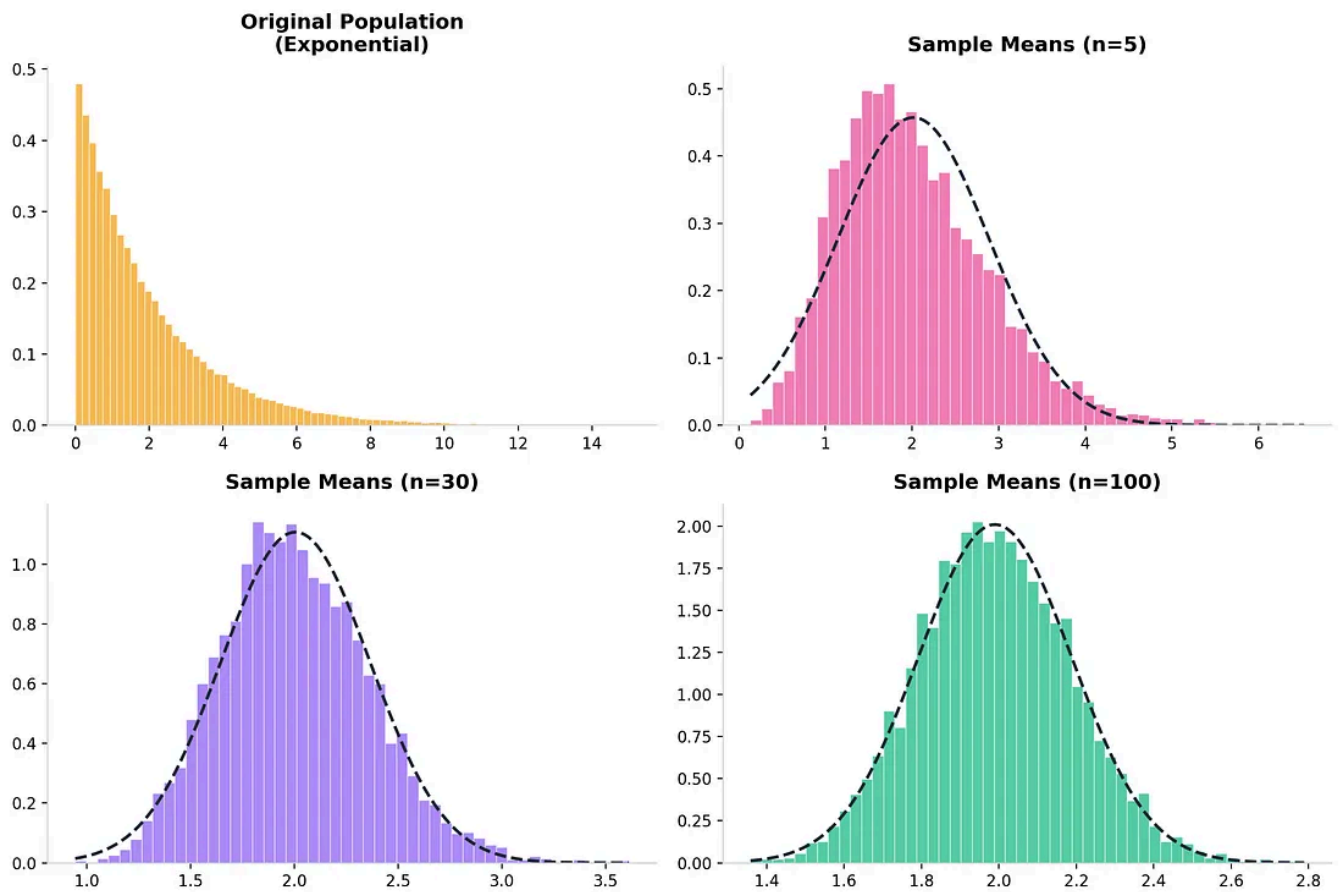
Here's the setup. Imagine you have a population where the data is shaped weirdly. Skewed to the right. Lumpy. Not a bell curve at all. Maybe it's the distribution of house prices in a city, where most houses are \$200K-\$400K but a few mansions push the tail out to \$10 million.

Now do this experiment. Pull a random sample of 30 houses. Calculate their average price. Write it down. Do it again. New sample, new average, write it down. Do it 1,000 times.

Those 1,000 averages? Plot them on a chart. They'll form a bell curve. Even though the original data was wildly skewed.

That is the Central Limit Theorem (CLT). No matter how weird the original population looks, the distribution of sample means will approach a normal distribution as the sample size gets big enough. "Big enough" is generally $n \geq 30$.

Central Limit Theorem in Action



Top-left is the ugly original data. Bottom-right is what happens when you take enough sample means. Pure bell curve. Every time.

Look at that transformation. The original population is an exponential distribution, completely lopsided. But by $n=30$, the sample means are beautifully normal. By $n=100$, it's textbook.

Why you should care. The CLT is the permission slip that lets us use normal distribution math (z-tables, t-tables, all the hypothesis testing formulas) even when our actual data isn't normally distributed. Without the CLT, we'd need every dataset to be perfectly bell-shaped before we could test anything. Real data is almost never bell-shaped. The CLT says "doesn't matter, the averages will be."

This is why $n \geq 30$ shows up as a magic number in statistics. It's the threshold where the CLT kicks in reliably.

Hypothesis Testing. Your Spam Filter Already Knows How to Do This.

Your email spam filter makes a decision about every incoming message. It starts with an assumption. “This email is probably legitimate.” Then it checks the evidence. Weird sender? Suspicious links? ALL CAPS SUBJECT LINE PROMISING FREE MONEY? If the evidence is strong enough, the filter overrides its default assumption and flags it as spam.

That is hypothesis testing. The entire framework.

You start with a default assumption (called the **null hypothesis**, H_0). You look at the evidence (your data). If the evidence is strong enough, you reject the default. If not, you stick with it.

Here’s the formal version.

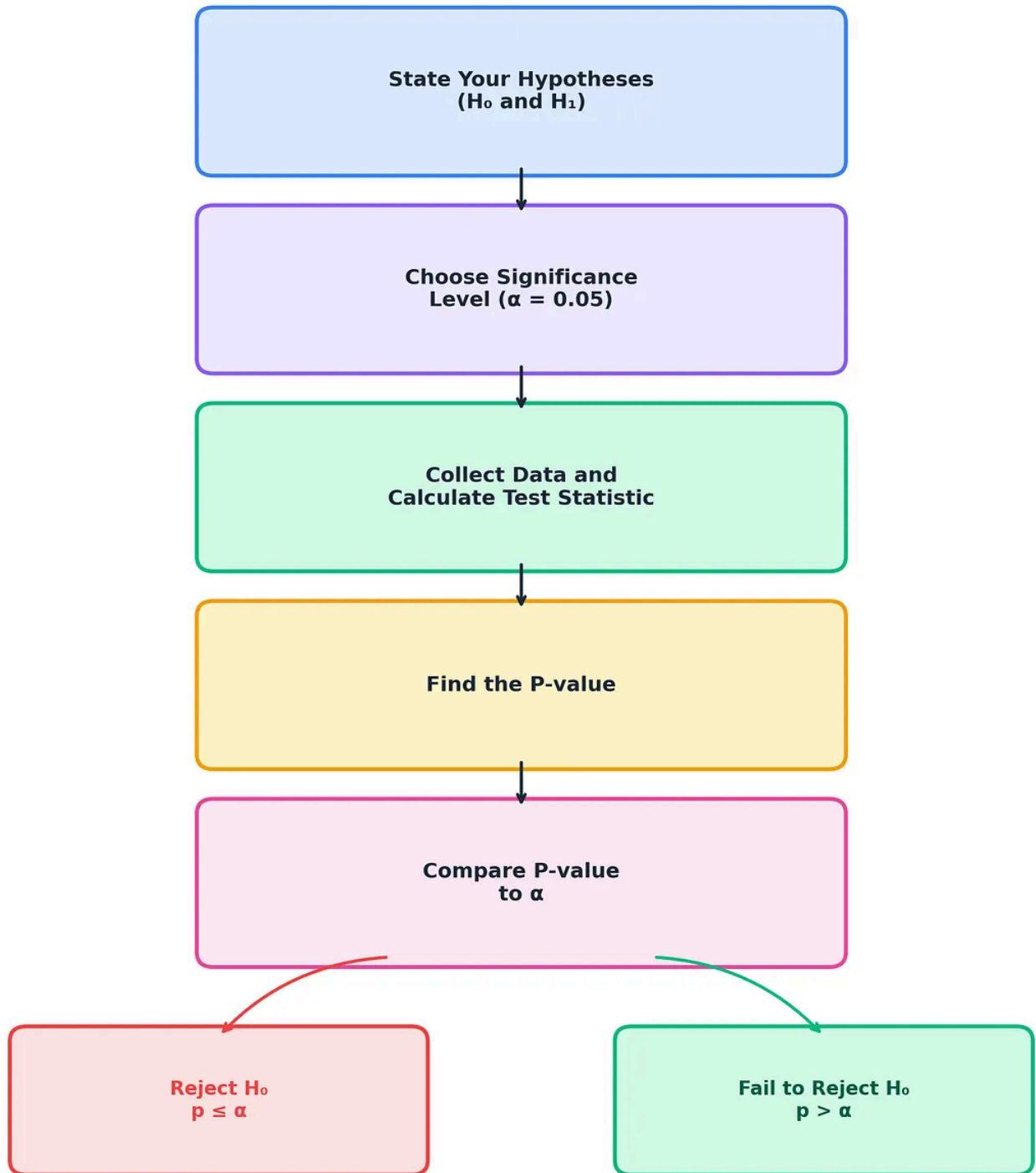
H_0 (**Null Hypothesis**) is always the boring claim. “Nothing new is happening. No difference exists. The drug doesn’t work. The new website design didn’t change anything.” H_0 is the status quo. The default. The “move along, nothing to see here.”

H_1 (**Alternative Hypothesis**) is the claim you’re actually interested in. “The drug DOES work. The new design DID increase clicks. The teaching method DID improve scores.” H_1 is what you’re hoping the data supports.

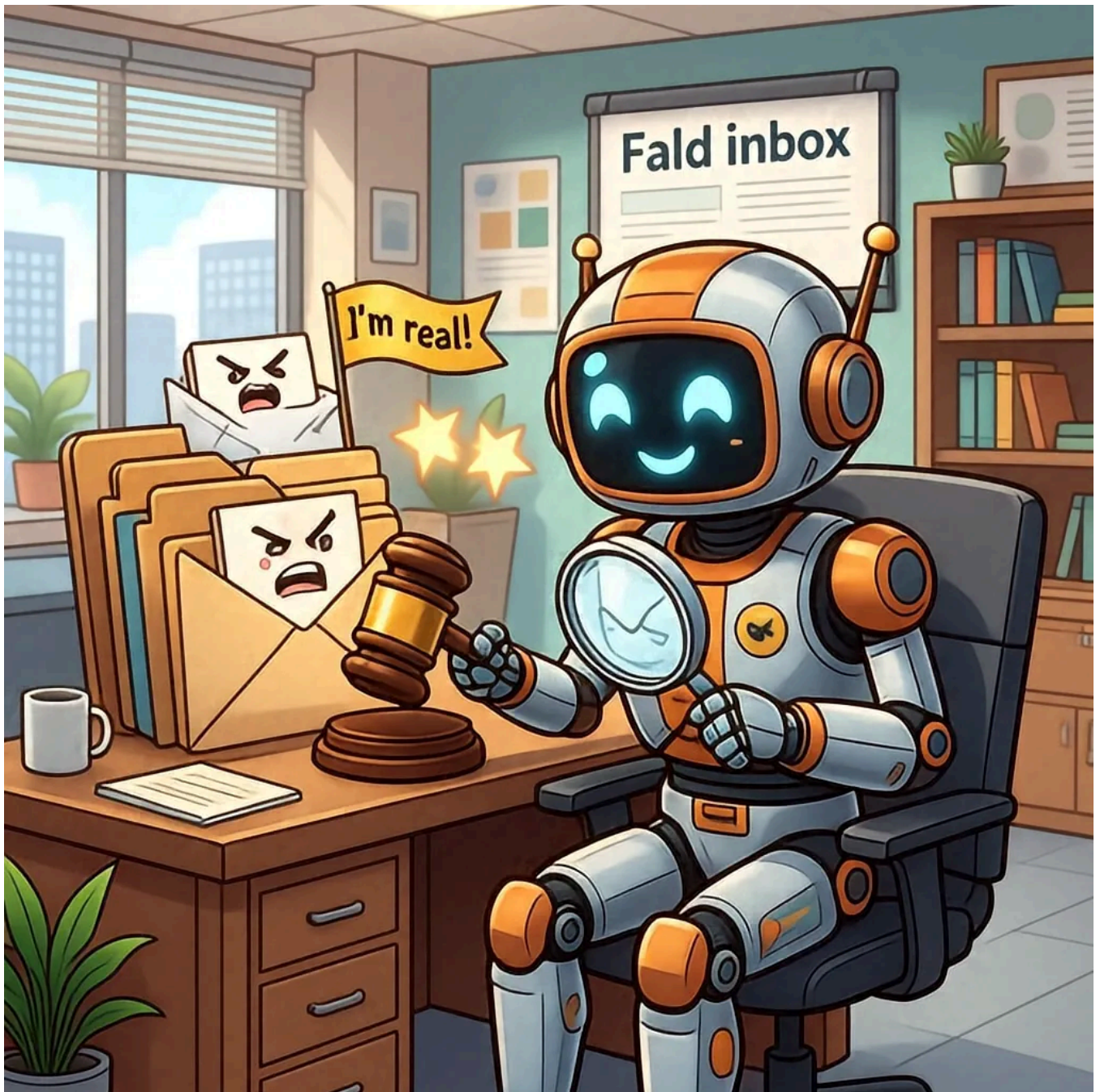
Your job is to collect data and ask one question. “Is this data so unlikely under H_0 that I should stop believing H_0 ?”

You never prove H_1 directly. You build a case against H_0 . If the case is strong enough, you reject H_0 , and H_1 wins by elimination. If the case is weak, you say “I don’t have enough evidence to reject H_0 .” Notice the language. You don’t say “ H_0 is true.” You say “I can’t reject it.” Big difference. Like a jury saying “not guilty” instead of “innocent.”

Hypothesis Testing Steps



Five steps. State your claim, set your threshold, collect data, calculate, decide. This flowchart is your new best friend.



Your spam filter has been doing hypothesis testing since 2004. You're just learning the vocabulary now.

Alpha. How Much Risk You're Willing to Accept.

Before looking at any data, you set a threshold. How much evidence is "enough"?

That threshold is called **alpha** (α), the significance level. The standard is $\alpha = 0.05$, which means 5%. But what does 5% actually mean here?

It means this. If absolutely nothing interesting is happening (H_0 is true), there's only a 5% chance your data would look this extreme purely by random luck. You're telling

the universe “I’m willing to be wrong 5% of the time when I call something significant.”

Why 0.05 specifically? Because a statistician named Ronald Fisher suggested it in the 1920s and the entire scientific world just... went with it. It’s not a law of physics. It’s a convention. Particle physicists use $\alpha = 0.0000003$ because they can’t afford false alarms when they’re claiming to have discovered new particles. Some social science researchers use $\alpha = 0.10$ for exploratory work.

The important thing is you set α BEFORE you see your data. Choosing it after is like throwing a dart and then drawing the bullseye around it.

P-Values. The Most Misunderstood Number in All of Science.

A 2025 study found that psychology papers reporting stronger p-values get published in better journals and receive more citations. This has created a perverse incentive where researchers, consciously or not, chase small p-values. Which makes it even more important that you actually understand what this number means.

So let’s be really precise.

A **p-value** is the probability of seeing results at least as extreme as yours, IF the null hypothesis is true.

Every word in that sentence is doing work. Let me unpack them one at a time.

“**IF the null hypothesis is true**” means the p-value lives in a hypothetical world where nothing interesting is happening. It’s asking “suppose the drug doesn’t work and we’re just looking at random noise, how often would we see data this dramatic?”

“**At least as extreme**” means not just your exact result, but anything even more surprising.

“**Probability**” means a number between 0 and 1. Closer to 0 means “this would almost never happen by accident.” Closer to 1 means “this could totally happen by accident.”

Let’s see this in a real scenario. You flip a coin 100 times and get 60 heads. Your null hypothesis is “the coin is fair” (50/50). Your p-value asks “if the coin really is fair, what’s the probability of getting 60 or more heads out of 100?” Turns out, it’s about

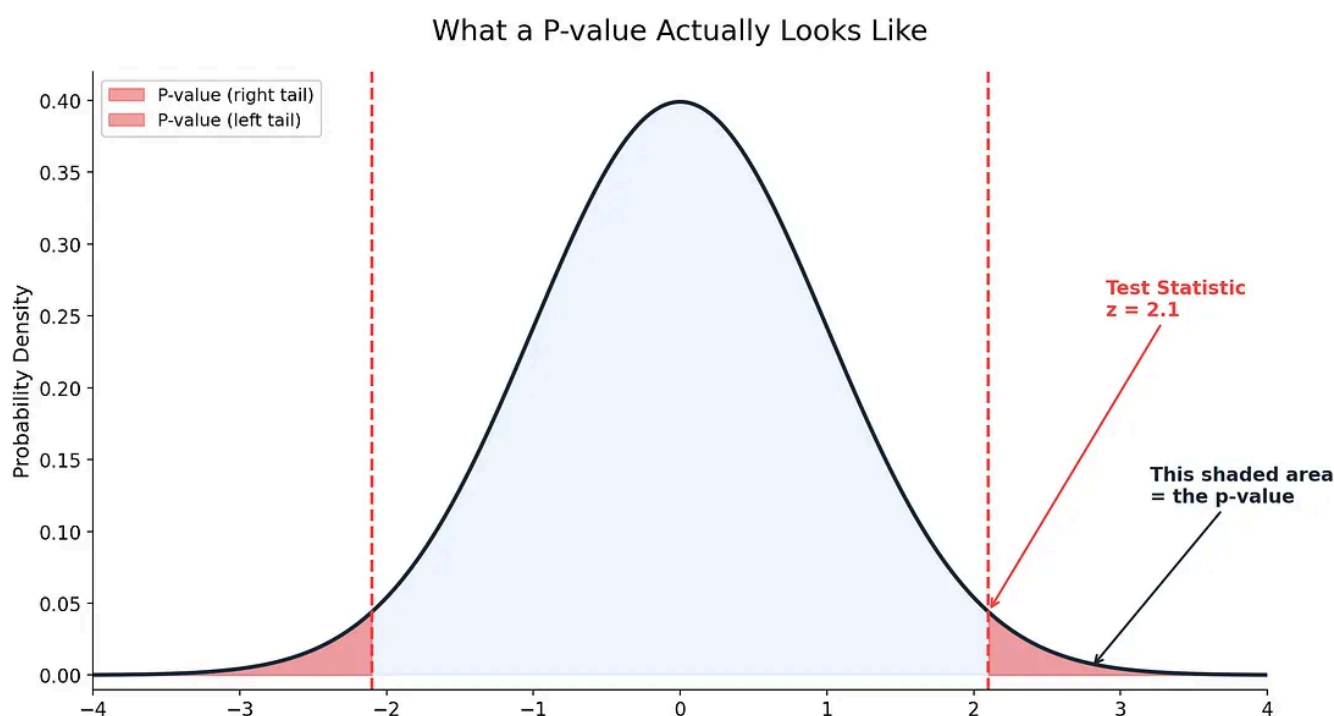
0.028. Under 3%. Pretty unlikely. So you'd probably reject H_0 and conclude the coin might be biased.

Now imagine you got 52 heads. The p-value for that is about 0.69. Getting 52 heads from a fair coin? Happens all the time. No reason to suspect anything.

The decision rule is simple.

If $p \leq \alpha$ (usually 0.05), reject H_0 . Your result is statistically significant.

If $p > \alpha$, fail to reject H_0 . Not enough evidence.



Those tiny shaded areas in the tails? That's your p-value. The smaller they get, the worse things look for H_0 .

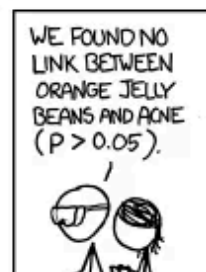
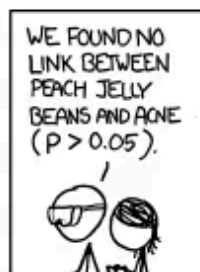
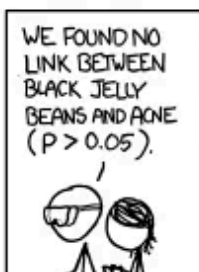
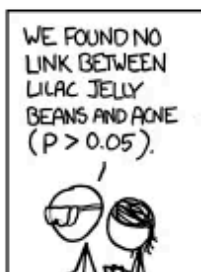
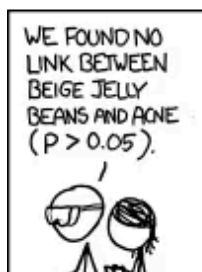
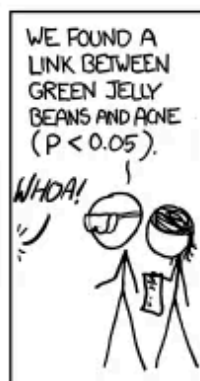
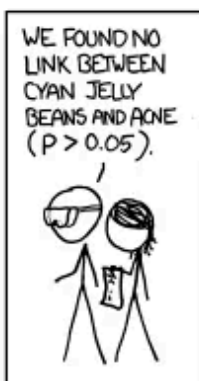
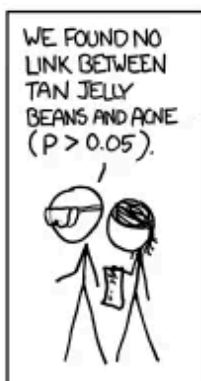
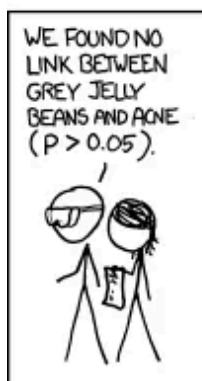
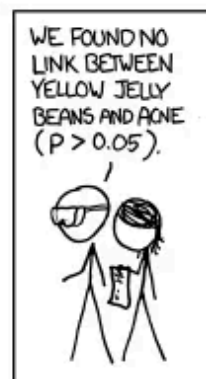
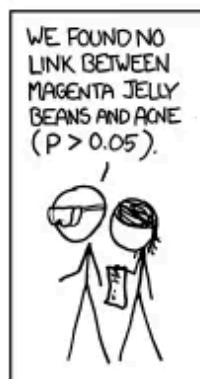
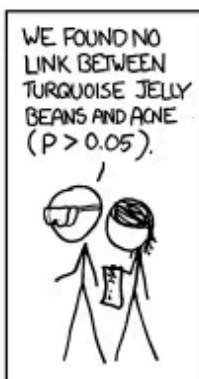
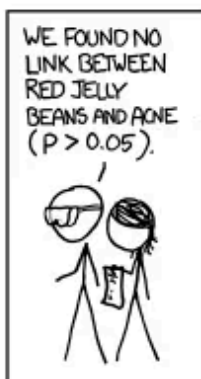
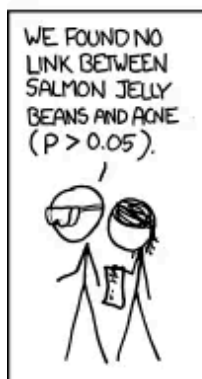
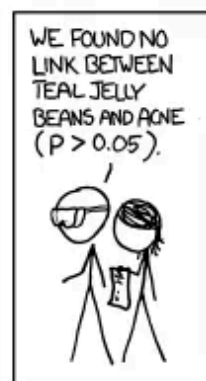
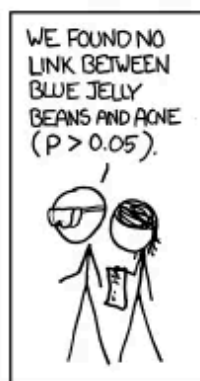
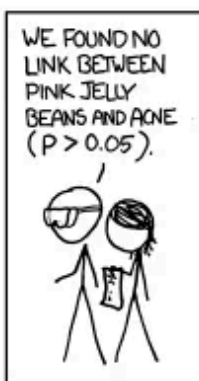
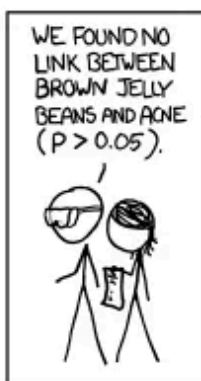
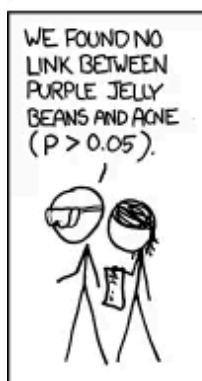
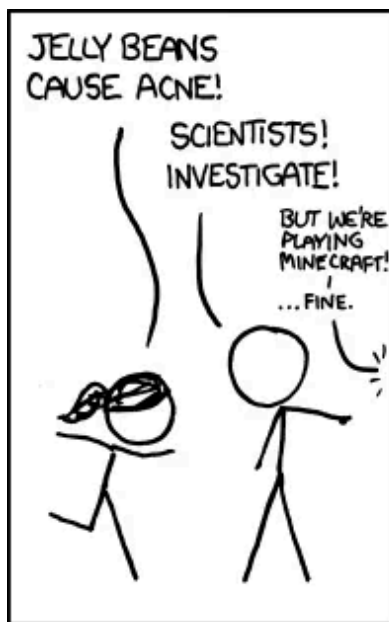
Now here's what a p-value is **NOT**, and getting this wrong is so common that the American Statistical Association published an official statement about it in 2016.

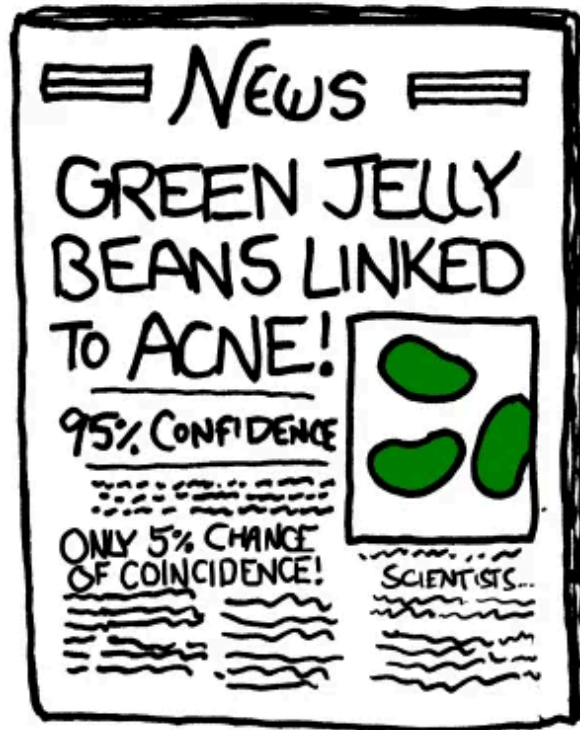
A p-value is not the probability that H_0 is true. Saying “ $p = 0.03$ means there's a 3% chance nothing is happening” is wrong. The p-value tells you about the data, not about the hypothesis.

A p-value is not the probability your result happened by chance. Close, but technically incorrect. It's the probability of data this extreme IF everything is due to chance.

A p-value does not measure how big or important the effect is. A drug that lowers blood pressure by 0.1 mmHg (medically useless) can have a p-value of 0.001 if you test 50,000 people. Significant doesn't mean important. More on this later.

A p-value of 0.04 is not fundamentally different from 0.06. Treating 0.05 as a cliff edge where one side is “real” and the other is “nothing” is one of the biggest problems in modern science.





The most famous comic in statistics. Twenty tests, one false positive, one published paper. This is exactly how p-hacking works.



0.049 and 0.051 walk into a bar. Only one of them gets published. That's the problem.

Type I and Type II Errors. The Two Flavors of Being Wrong.

Every time you make a decision based on a hypothesis test, you can be wrong in exactly two ways.

Type I Error. The False Alarm.

You rejected H_0 , but H_0 was actually true. You said "something's happening!" when nothing was happening. Your spam filter marked your mom's email as junk. The fire alarm goes off because someone burned toast.

The probability of this equals your alpha. If $\alpha = 0.05$, you'll trigger a false alarm about 5% of the time. That's the price of admission. You chose it.

Type II Error. The Missed Signal.

You failed to reject H_0 , but H_0 was actually wrong. Something real was happening and you missed it. Your spam filter let a phishing email through. You slept through a real fire because you assumed the alarm was another toast situation.

The probability of this is called β (beta). And $1 - \beta$ is called **power** (more on that later).

Type I and Type II Errors

	Reject H_0	Fail to Reject H_0
H_0 is Actually TRUE	TYPE I ERROR (α) "False Alarm"	CORRECT ✓ "True Negative"
H_0 is Actually FALSE	CORRECT (Power = $1 - \beta$) "True Positive"	TYPE II ERROR (β) "Missed Detection"

Four possible outcomes, two correct, two wrong. The colors tell you which is which.

Which one is worse? Depends on the context.

In a **medical screening**, a Type II error (missing an actual disease) can kill someone. You'd rather have a few false alarms than miss a cancer diagnosis. So you set α higher (maybe 0.10) to catch more true positives, accepting more false positives as the tradeoff.

In a **criminal trial**, a Type I error (convicting an innocent person) is considered worse. The system is designed to let some guilty people walk free rather than imprison innocents. That's why the standard is "beyond a reasonable doubt" (a very low α).

In **drug approval**, the FDA uses $\alpha = 0.05$ because approving a useless drug (Type I) wastes money and gives false hope, while rejecting a working drug (Type II) means patients miss a treatment. Both are bad. The 0.05 is a compromise.

The insight here is that you can't minimize both errors at the same time with a fixed sample size. Lowering α (fewer false alarms) automatically increases β (more missed signals). The only way to reduce both is to collect more data.



The fire alarm dilemma. Too sensitive and it screams at toast. Too quiet and it sleeps through a real fire. That's the alpha-beta tradeoff.

The Z-Test. The One You Learn First but Use Last.

Time for actual tests. Let's start with the z-test because it's the foundation everything else is built on.

A **z-test** compares your sample mean to a known population mean when you know the population standard deviation (σ). That “when you know σ ” part is crucial and also the reason you'll rarely use this test in real life. Knowing the population standard deviation requires either massive historical datasets or published industry specs. It's like having the answer key to an exam. Useful when you have it. Almost never available.

But the logic of the z-test is the logic of every test that follows. Understand this one and the rest are variations on the same theme.

The formula.

$$z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

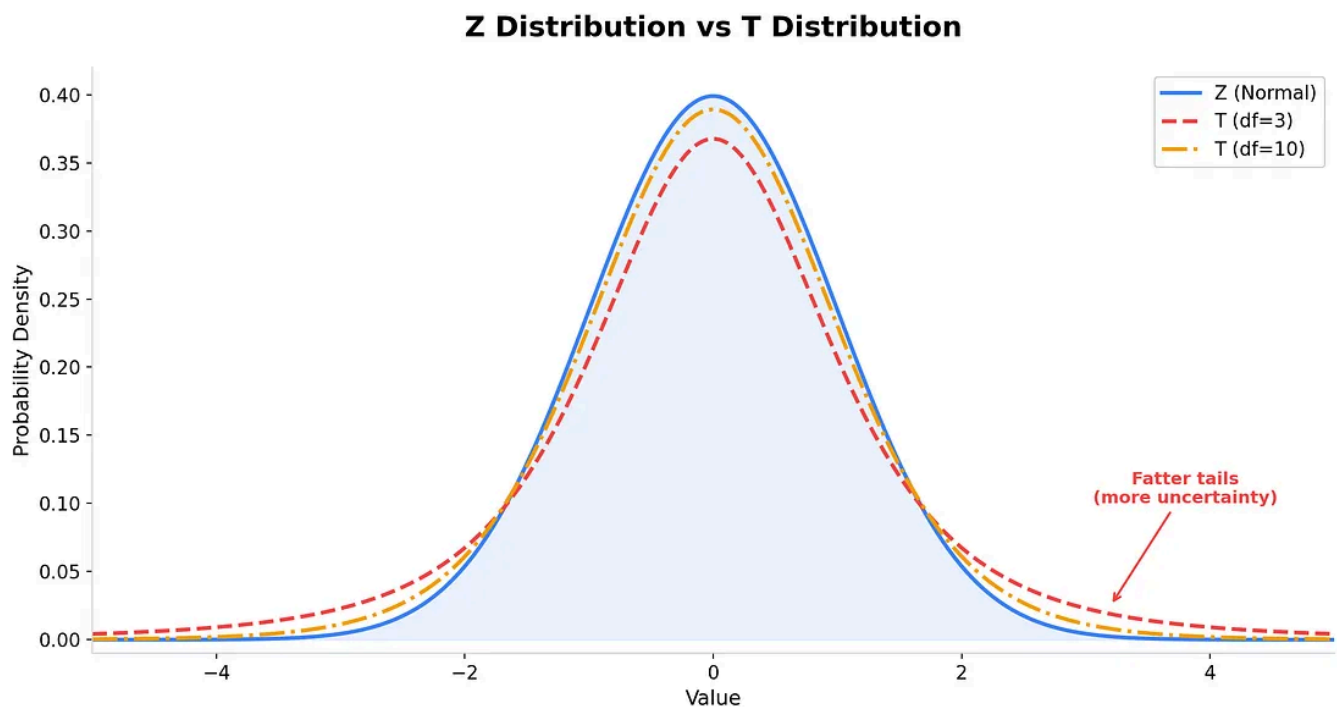
Looks intimidating. It isn't. Let me translate each piece.

The top part ($\bar{x} - \mu$) is just the distance between your sample average and the population average. “How far off is my sample?”

The bottom part ($\sigma / \sqrt{n}$) is called the **standard error**. It measures how much sample averages naturally bounce around due to random chance. Think of it as the “noise level.” Dividing σ by the square root of n makes sense because bigger samples have less noise (more data = more stable average).

The z-value itself tells you how many standard errors your sample is away from the population mean. A z of 1 means “one standard error away.” A z of 2 means “two standard errors away,” and that's already unusual. A z of 3 means “three standard errors away,” and now you should be suspicious.

Why? Because the 68–95–99.7 rule tells us that values beyond 2 standard deviations happen less than 5% of the time by chance. Beyond 3? Less than 0.3%.



The blue line is the z-distribution (standard normal). The red dashed line is the t-distribution with only 3 data points. See how much fatter the tails are? That extra width accounts for the uncertainty when your sample is tiny.

Now let's see the whole thing in Python.

```
# numpy handles the math, scipy.stats has the test functions
# These two libraries power 90% of statistics work in Python
import numpy as np
from scipy import stats
#...
# SCENARIO: A battery factory claims their batteries last
# 500 hours on average, with a known sigma of 30 hours.
# "Known" because they've been manufacturing for 20 years
# and have data on millions of batteries.
# We're the quality inspector. We grabbed 36 batteries off
# the line and tested them. Do they actually last 500 hours?
population_mean = 500 # what the factory claims (mu)
population_std = 30   # known from 20 years of data (sigma)
#...
# Our 36 test results in hours
# 36 is our sample size (n). It's > 30, so CLT is happy.
sample = [510, 485, 520, 505, 515, 498, 525, 490, 512,
          508, 530, 495, 518, 503, 522, 487, 510, 505,
          515, 502, 528, 493, 510, 507, 520, 499, 513,
          506, 518, 497, 515, 504, 522, 509, 511, 503]
#...
```

```

# Step 1: Calculate the sample mean (x-bar)
# np.mean() adds everything up and divides by count
sample_mean = np.mean(sample)
n = len(sample)  # should be 36
#...
# Step 2: Calculate the z-statistic
# Top: how far our sample mean is from the claimed mean
# Bottom: the standard error (sigma / sqrt of n)
# The standard error shrinks as n grows, which is why
# bigger samples give you more precise estimates
z_stat = (sample_mean - population_mean) / (population_std / np.sqrt(n))
#...
# Step 3: Get the p-value
# stats.norm.cdf(z) gives the area under the curve LEFT of z
# For a two-tailed test (we're checking "different from 500"
# in either direction), we need both tails
# abs() because the tail area is the same whether z is
# positive or negative (the curve is symmetric)
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))
#...
# Step 4: Make the call
alpha = 0.05
print(f"Sample mean: {sample_mean:.2f} hours")
print(f"Z-statistic: {z_stat:.4f}")
print(f"P-value: {p_value:.4f}")
if p_value <= alpha:
    print("REJECT H0 → batteries are NOT lasting 500 hours")
else:
    print("FAIL TO REJECT H0 → no evidence against the claim")

```

What each step actually did. Step 1 found the center of our sample. Step 2 measured how far that center was from the factory's claim, adjusted for noise. Step 3 asked “if the factory is telling the truth, how likely is it we'd see a sample this extreme?” Step 4 compared that likelihood to our tolerance for being wrong ($\alpha = 0.05$).

Why not always use the z-test? Because you need to know σ , and you almost never do. When was the last time someone handed you the exact standard deviation of an entire population? In the real world, you estimate σ from your sample data. And the moment you do that, you need the t-test.

The T-Test. Born in a Brewery, Used Everywhere.

Here's one of my favorite stories in all of science.

In 1908, a chemist named William Sealy Gosset was working at the Guinness brewery in Dublin. His job was quality control. He needed to test whether small batches of barley met standards, but his samples were tiny. The statistical methods available at the time (built on the normal distribution) required large samples, and they gave unreliable results with small ones. Nobody had solved this problem.

So Gosset solved it himself. He invented the t-test. But Guinness had a strict policy against employees publishing research. They didn't want competitors learning about their scientific methods. So Gosset published his work under the pseudonym "Student." His identity wasn't publicly revealed until his death in 1937.

That is why it's called **Student's t-test**. Not because it's for students. Because a brewery chemist had to sneak his most important work into a journal under a fake name.

What makes the t-test different from the z-test? One thing. Instead of using the known population σ , you estimate it from your sample using s (the sample standard deviation). This introduces extra uncertainty, and the t-distribution accounts for that uncertainty by having fatter tails than the normal distribution. Fatter tails mean you need stronger evidence to reject H_0 , which makes sense. Less certainty about the spread $\rightarrow$ need to be more careful before making claims.

As your sample size grows, s gets closer and closer to σ , and the t-distribution looks more and more like the normal distribution. Past $n \approx 30$, they're practically identical.

There are three types of t-test. Each asks a different question. Let me walk through all three using one scenario so you can see how they connect.

One-Sample T-Test. "Is this group different from a target?"

The scenario. You're a school administrator. The national average exam score is 75. You want to know if your school's students are performing differently. You grab 25 students' scores.

```
from scipy import stats
import numpy as np
#...
# 25 exam scores from our school
# These are real-ish numbers, not cherry-picked
```

```

scores = [82, 78, 85, 90, 73, 88, 76, 81, 84, 79,
          87, 91, 77, 83, 86, 80, 74, 89, 82, 85,
          78, 84, 88, 76, 81]

#...
# H0: our school's average = 75 (same as national)
# H1: our school's average ≠ 75 (different)
# This is two-tailed because we're not specifying which
# direction the difference goes
#...
# ttest_1samp takes two arguments:
# 1) your data (the 25 scores)
# 2) popmean: the value you're comparing against (75)
# It returns two things: the t-statistic and the p-value
# scipy does ALL the math internally (calculates the sample
# mean, sample std, standard error, degrees of freedom)
t_stat, p_value = stats.ttest_1samp(scores, popmean=75)
#...
print(f"Sample mean: {np.mean(scores):.2f}")
print(f"Sample std: {np.std(scores, ddof=1):.2f}")
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4f}")
print(f"df: {len(scores) - 1}")
#...
# ddof=1 in np.std() means "sample standard deviation"
# ddof stands for "delta degrees of freedom"
# We use ddof=1 because we're estimating from a sample,
# not calculating for an entire population
#...
alpha = 0.05
if p_value <= alpha:
    print("→ REJECT H0: our school differs from national avg")
else:
    print("→ FAIL TO REJECT: no significant difference found")

```

When to use this. You have one group of data and one specific number to compare it against.

Independent Two-Sample T-Test. “Are these two groups different from each other?”

Same school, new question. You now want to compare two teaching methods. Class A used traditional lectures. Class B used an interactive method. Different students in each class. You want to know if one method produced better scores.

```

from scipy import stats
import numpy as np
#...
# Class A: 20 students, traditional lectures
class_a = [72, 78, 81, 85, 74, 79, 83, 77, 80, 76,
           82, 75, 79, 84, 73, 81, 78, 80, 76, 83]
#...
# Class B: 20 students, interactive method
class_b = [85, 88, 82, 90, 87, 84, 91, 86, 89, 83,
           88, 85, 87, 92, 84, 86, 90, 88, 85, 87]
#...
# H0: mean(A) = mean(B) (no difference between methods)
# H1: mean(A) ≠ mean(B) (one method is better)
#...
# ttest_ind = independent samples t-test
# "Independent" because the students in class A are
# completely different people from the students in class B
# There's no overlap, no pairing, no connection
t_stat, p_value = stats.ttest_ind(class_a, class_b)
#...
print(f"Class A mean: {np.mean(class_a):.2f}")
print(f"Class B mean: {np.mean(class_b):.2f}")
print(f"Difference: {np.mean(class_b) - np.mean(class_a):.2f}")
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.6f}")
#...
alpha = 0.05
if p_value <= alpha:
    print("→ REJECT H0: the methods produce different results")
else:
    print("→ FAIL TO REJECT: no significant difference found")

```

When to use this. You have two separate groups of different people/items and want to compare them.

Paired T-Test. “Did something change in the same group?”

Same school, yet another question. You pick 15 students. You test them before using a new study app, then test them again after 4 weeks with the app. Same students both times. You want to know if the app helped.

This is different from the two-sample test because you’re measuring the same humans twice. Each “before” score has a partner “after” score.

```

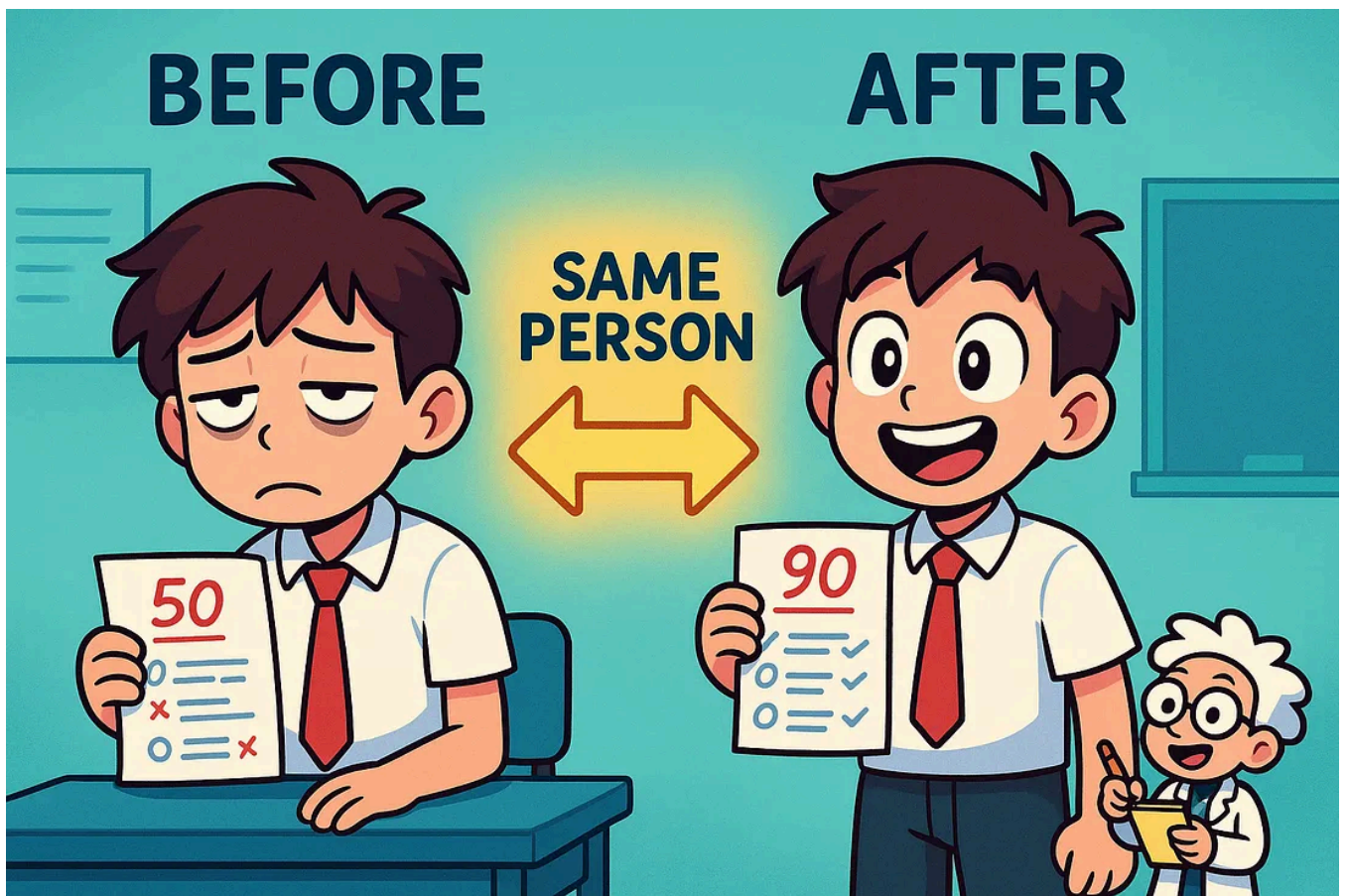
from scipy import stats
import numpy as np
#...
# SAME 15 students, measured TWICE
# This pairing is why it's called a "paired" test
before = [65, 70, 72, 68, 74, 66, 71, 69, 73, 67,
          70, 72, 68, 75, 71]
after  = [75, 78, 80, 74, 82, 73, 79, 77, 81, 74,
          78, 80, 76, 83, 79]

#...
# H0: the app made no difference (mean difference = 0)
# H1: the app made a difference (mean difference ≠ 0)
#...
# ttest_rel = related (paired) samples t-test
# "Related" because each before has a matching after
# Under the hood, it calculates the difference for each
# pair and then does a one-sample t-test on those
# differences, testing if the mean difference = 0
t_stat, p_value = stats.ttest_rel(before, after)
#...
# Let's also look at the individual improvements
diffs = np.array(after) - np.array(before)
print(f"Before mean:      {np.mean(before):.2f}")
print(f"After mean:      {np.mean(after):.2f}")
print(f"Mean improvement: {np.mean(diffs):.2f} points")
print(f"T-statistic:      {t_stat:.4f}")
print(f"P-value:         {p_value:.6f}")
#...
alpha = 0.05
if p_value <= alpha:
    print("→ REJECT H0: the app made a significant difference")
else:
    print("→ FAIL TO REJECT: improvement was not significant")

```

Why not just use the two-sample test here? Because the paired test is more powerful when you can pair observations. Student #3 might be a naturally weak test-taker. That weakness appears in both their before AND after score. The paired test looks at the *difference within each pair*, which cancels out individual baseline differences. The two-sample test ignores the pairing and treats them as strangers, losing information.

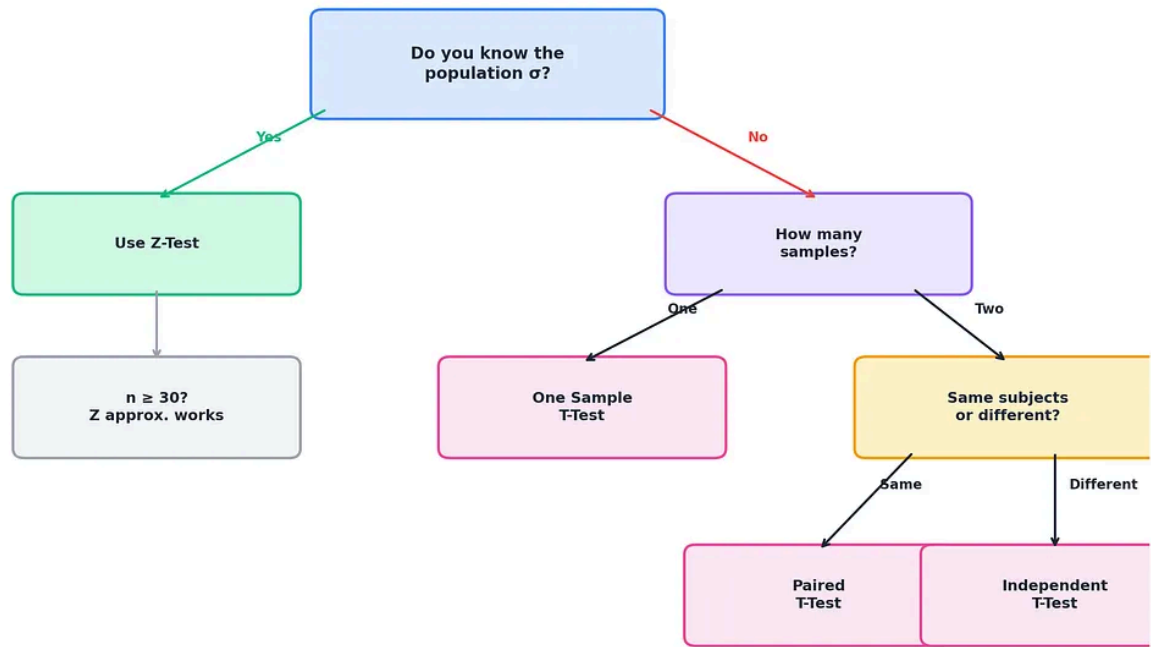
If your data has a natural pairing (same person, same machine, same city measured twice), always use the paired test.



Same student, two scores, one question. Did something actually change or are we kidding ourselves?

When to Use What. The Only Decision Tree You Need.

Which Statistical Test Should You Use?



Start at the top. Follow the arrows. Arrive at the right test. Print this out.

Here's the same thing as a quick-reference table.

Situation	Use This Test
Know σ , large sample	Z-test
Don't know σ , one group	One-sample t-test
Don't know σ , two groups	Independent two-sample t-test
Same subjects, two times	Paired t-test

Honestly? If you're ever unsure, just use a t-test. If you know σ (you probably don't), use a z-test. If n is above 30, the results will be nearly identical either way because the t-distribution converges to the normal distribution. The t-test is the Swiss Army knife. The z-test is the textbook exercise.

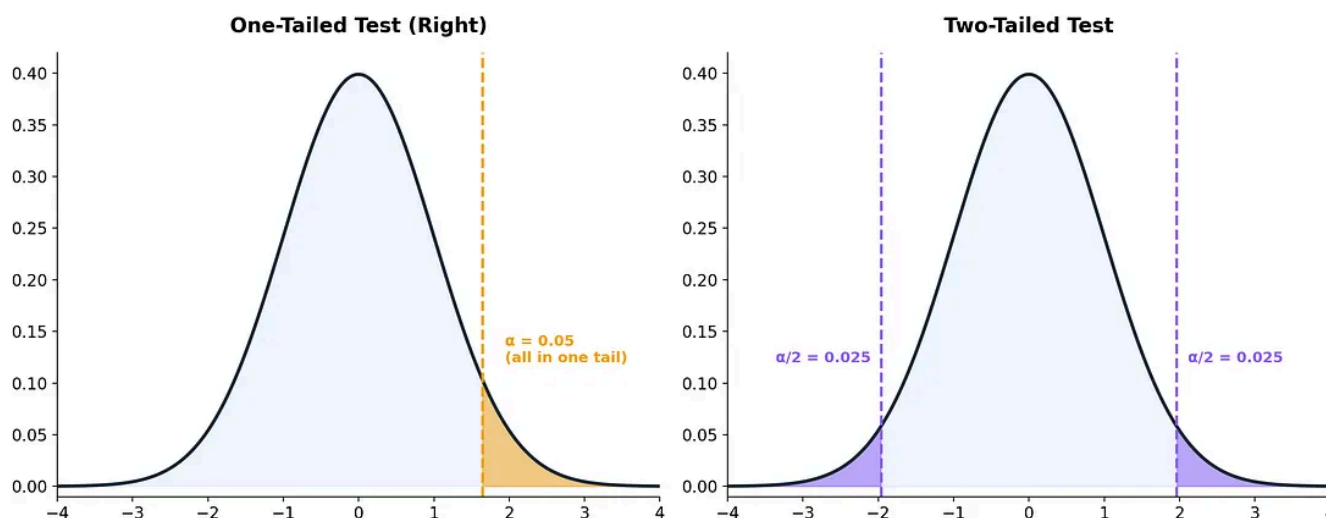
One-Tailed vs Two-Tailed. Quick but Important.

When you set up H_1 , you're making a directional choice.

Two-tailed. "Is the mean DIFFERENT from X?" Could be higher or lower. You don't care which direction, just whether it's different. The rejection region (that shaded area where you reject H_0) is split between both ends of the curve. Each tail gets $\alpha/2$.

One-tailed. "Is the mean GREATER than X?" or "Is the mean LESS than X?" You have a specific direction. All of α goes into one tail, making it easier to detect an effect in that direction.

One-Tailed vs Two-Tailed Tests



Left puts all the rejection power in one tail. Right splits it evenly. Same total area, very different sensitivity.

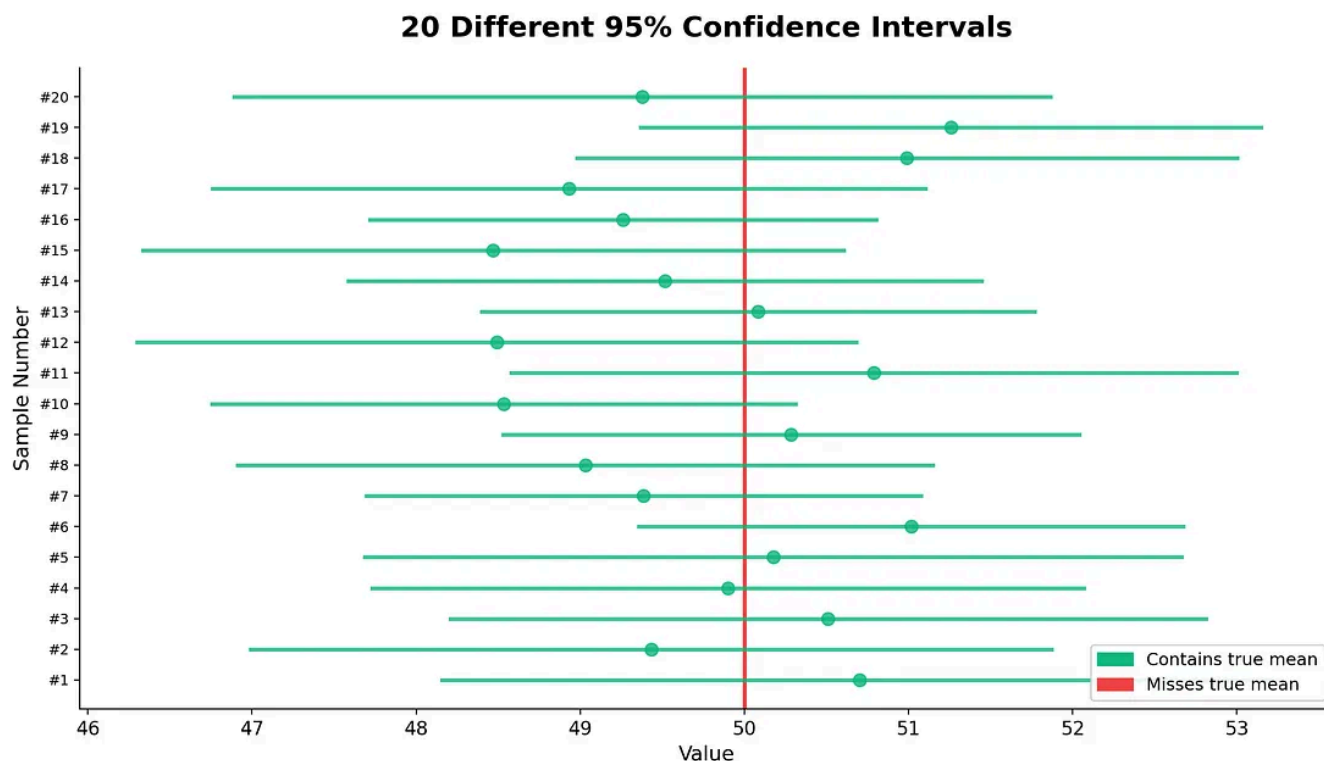
Rule of thumb. Default to two-tailed unless you have a strong, pre-existing reason to expect a specific direction. One-tailed gives more power in one direction but completely blinds you to effects in the other. If a drug could either raise or lower blood pressure, you want two tails. If its mechanism literally cannot raise blood pressure, one tail might be justified.

Confidence Intervals. The Range Version of Your Answer.

A hypothesis test gives you a yes/no answer. A confidence interval gives you a range. Both come from the same math, but the interval is often more useful.

A **95% confidence interval** is a range of values constructed so that if you repeated the experiment 100 times, about 95 of those intervals would contain the true population parameter.

Critical distinction. It does NOT mean “there’s a 95% chance the true value is inside this interval.” The true value is fixed. It’s either in there or it isn’t. The 95% describes the method, not any single result.



Twenty experiments, twenty intervals. The green ones caught the true mean. The red ones missed. Over time, about 95% will hit.

```
import numpy as np
from scipy import stats
#...
# Same school exam scores
scores = [82, 78, 85, 90, 73, 88, 76, 81, 84, 79,
          87, 91, 77, 83, 86, 80, 74, 89, 82, 85,
          78, 84, 88, 76, 81]
#...
sample_mean = np.mean(scores)
n = len(scores)
#...
# Standard error = sample std / sqrt(n)
# ddof=1 because we're working with a sample, not population
# sem() is a shortcut that does exactly this calculation
se = stats.sem(scores)
#...
```

```
# stats.t.interval builds the confidence interval
# Arguments:
#   confidence level (0.95 for 95%)
#   df: degrees of freedom (n - 1)
#   loc: center of interval (our sample mean)
#   scale: standard error (how much the mean bounces around)
ci = stats.t.interval(0.95, df=n-1, loc=sample_mean, scale=se)
#...
print(f"Sample mean: {sample_mean:.2f}")
print(f"95% CI: ({ci[0]:.2f}, {ci[1]:.2f})")
# This means we're 95% confident the true population
# mean falls somewhere in this range
```

Connection to hypothesis testing. If a 95% CI doesn't contain the value from your null hypothesis, then a two-tailed test at $\alpha = 0.05$ would reject H_0 . They're the same math expressed differently. The CI just gives you more information because it tells you the plausible range, not just "significant or not."

Degrees of Freedom. Fifteen Seconds, I Promise.

If you have 10 data points and you know their mean, only 9 of them can be anything they want. The 10th is locked because it has to make the total work out. Those 9 "free" values are your degrees of freedom.

For a one-sample t-test, $df = n - 1$. For an independent two-sample t-test, $df \approx n_1 + n_2 - 2$. For a paired t-test, $df = n_{\text{pairs}} - 1$.

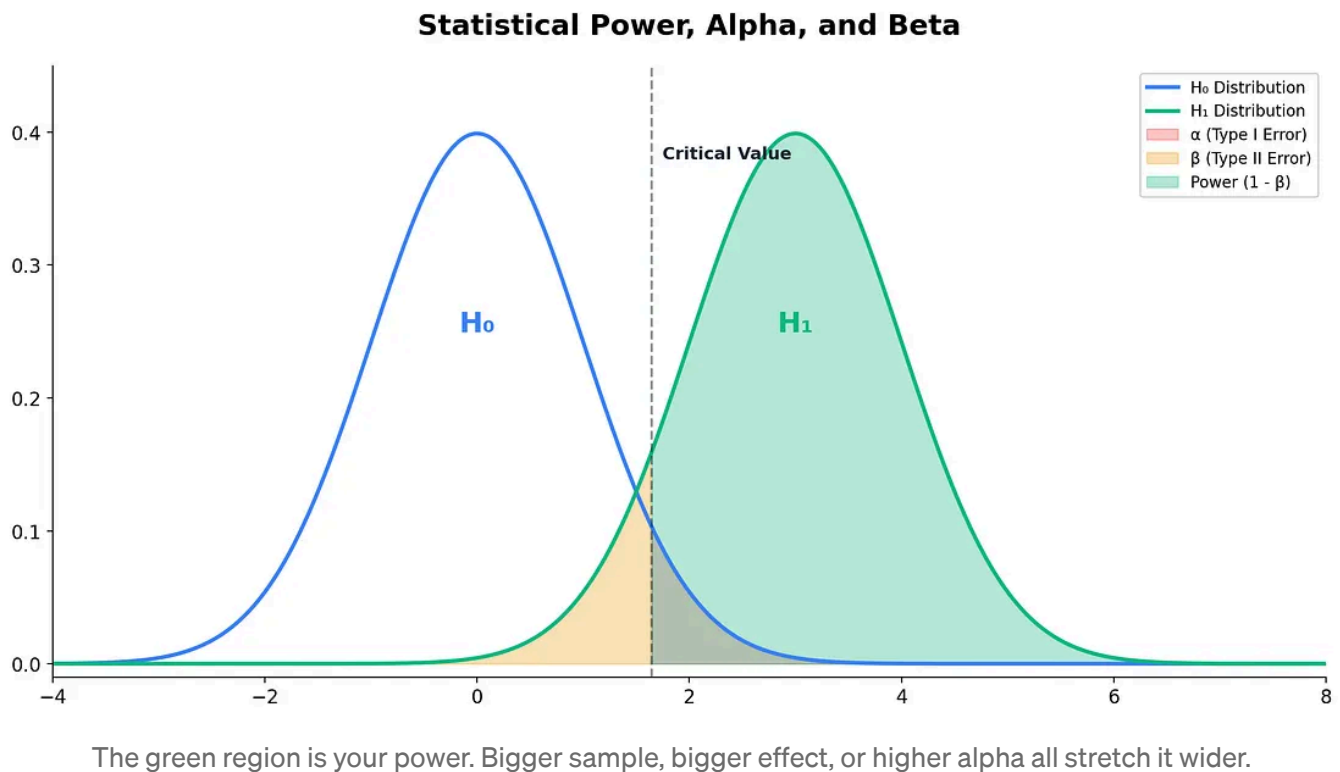
Lower df means fatter tails on the t-distribution, which means you need stronger evidence to reach significance. Higher df means the t-distribution looks more like the z-distribution. That's it. That's degrees of freedom.

Statistical Power and Effect Size. Why Smart Studies Still Fail.

In 2015, a massive research project tried to replicate 100 published psychology studies. Only 36% produced significant results the second time. This is the **replication crisis**, and a big part of the reason is that many original studies were underpowered. They didn't have enough data to reliably detect the effects they were looking for.

Statistical power is the probability that your test will correctly reject H_0 when something real is going on. $\text{Power} = 1 - \beta$.

A power of 0.80 (the standard benchmark) means an 80% chance of catching a real effect. A power of 0.50 means you're basically flipping a coin on whether you'll find what's actually there.



Four things increase power.

More data. Bigger n = more precision = more power.

Bigger effect. Larger real differences are easier to spot than tiny ones.

Higher alpha. Accepting more false alarm risk ($\alpha = 0.10$ instead of 0.05) gives you more power, but at a cost.

Less noise. Lower standard deviation in your data makes signals clearer.

Now, **effect size** tells you HOW BIG the difference is, separate from whether it's "statistically significant." This is where things get practical.

Cohen's d is the most common measure for comparing means.

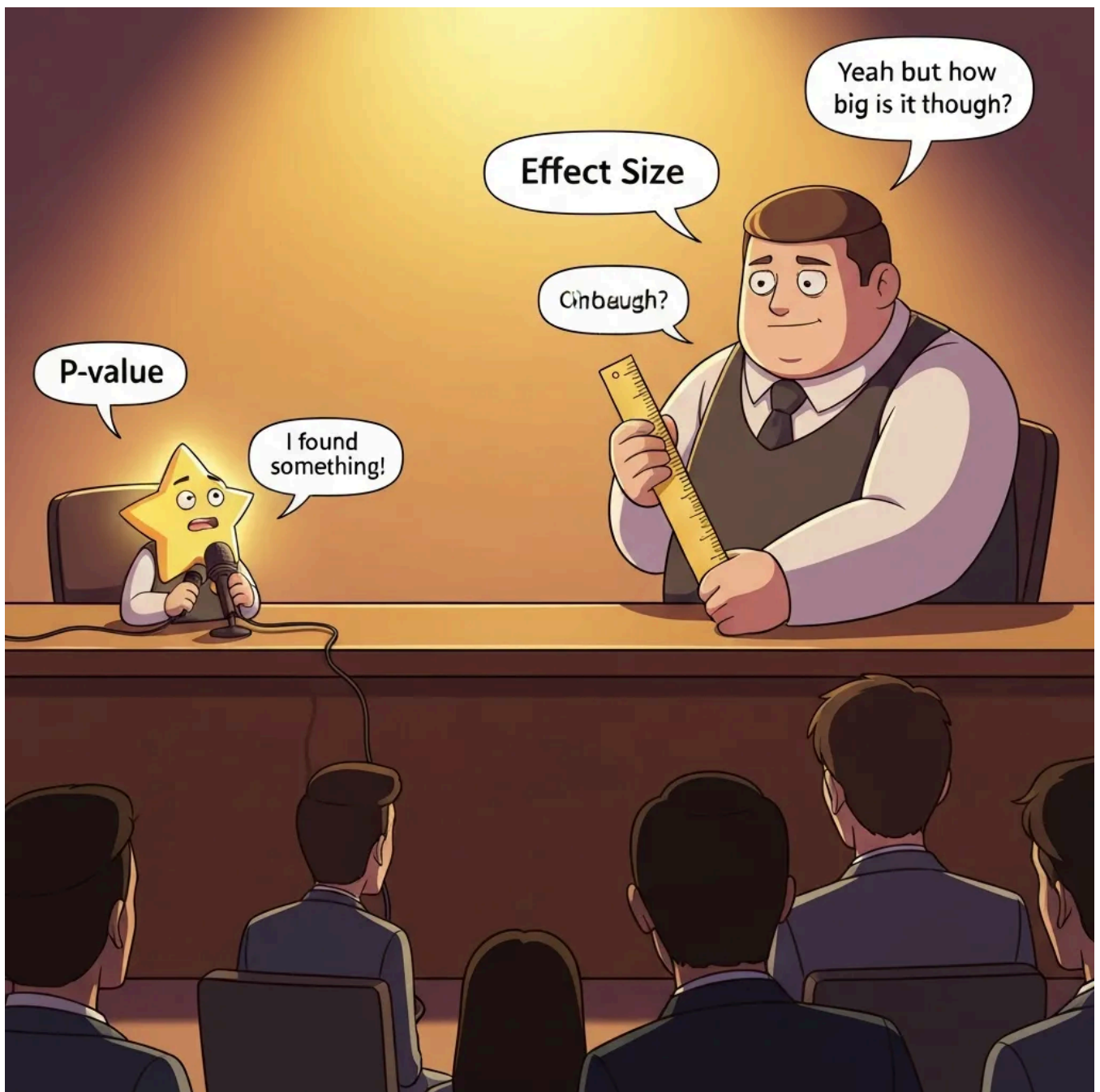
$$d = (\text{mean}_1 - \text{mean}_2) / \text{pooled_standard_deviation}$$

Cohen's rough guide. $d = 0.2$ is small (barely noticeable), $d = 0.5$ is medium (noticeable), $d = 0.8$ is large (hard to miss).

Here's why this matters. With 50,000 participants, you can get a p-value of 0.001 for a difference of 0.3 IQ points between two groups. Statistically significant? Yes. Practically meaningful? Absolutely not. Effect size catches what p-values miss.

A study published in a top journal with $p < 0.001$ and $d = 0.04$ discovered a "real" effect that's so small nobody could ever notice it. A study rejected with $p = 0.07$ but $d = 0.9$ might have found something genuinely important but didn't have enough data to clear the significance bar.

Always report both. P-value without effect size is like saying "something changed" without saying how much. You wouldn't accept "the patient's condition is different" from a doctor without asking "better or worse, and by how much?"



P-value says "something is there." Effect size says "yeah, but is it worth caring about?"

The Mistakes That'll Sink You. Let Me Save You the Embarrassment.

I've watched people who should know better make these. Some of them were me.

Saying "we accepted H_0 ." You never accept H_0 . You fail to reject it. The difference isn't pedantic. "We found no evidence against H_0 " is very different from "we proved H_0 is true." An absence of evidence is not evidence of absence. Maybe your sample was just too small.

P-hacking. Running test after test on the same data, adjusting variables, removing outliers, switching between one-tailed and two-tailed, until something crosses $p < 0.05$. If you test 20 independent things at $\alpha = 0.05$, on average one will be “significant” by pure luck. A 2025 study found that fraudulent science publications are now growing faster than legitimate ones. P-hacking is part of that story.

Using the wrong test for your data. Paired data with an independent test throws away information. A z-test when you don’t know σ gives overconfident results. Small sample without checking normality can produce garbage.

Ignoring effect size. Already covered, but it’s so common that it deserves a second mention. Significant is not a synonym for important.

Setting alpha after peeking at results. If your p-value is 0.048 and you retroactively set $\alpha = 0.05$, that’s not science. You drew the bullseye after the throw.

Confusing statistical significance with real-world significance. A significant result means “unlikely to be pure chance.” It doesn’t mean “big enough to matter” or “clinically relevant” or “worth changing your business strategy over.” Always ask “significant, but so what?”

The Quick Reference Card

Concept	One-Line Translation
Statistical Inference	Guessing the whole from a piece
Population	Everyone you care about
Sample	The slice you actually measured
Standard Deviation	Average distance from the average
Normal Distribution	The bell curve. Data piles in the middle, thins at the edges.
CLT	Sample means form a bell curve no matter what the raw data looks like (if $n \geq 30$)
H0	"Nothing interesting here"
H1	"Something IS going on"
Alpha	Your false alarm budget (usually 5%)
P-value	How weird your data is IF H0 is true
Type I Error	False alarm. Rejected a true H0.

Type II Error	Missed it. Kept a false H_0 .	
Z-Test	Know sigma? Use this. (Rare.)	
T-Test	Don't know sigma? Use this. (Common.)	
Confidence Interval	Plausible range for the true value	
Degrees of Freedom	$n - 1$. Affects the t-curve shape.	
Power	Chance of catching a real effect	
Effect Size	HOW BIG the difference actually is	
+-----+-----+		

Now What?

But reading about statistics and doing statistics are different skills. The jump from one to the other is smaller than you think, though. Open a Jupyter notebook. Find a dataset on Kaggle (search “beginner friendly dataset”). Pick a question. “Is the average price of houses in this neighborhood different from the city average?” Pick a test. Run it. Mess it up. Fix it.

That’s how every person who’s good at this actually learned.

If you remember just three things from this entire article, let them be these. First, p-values tell you how surprising your data is, not how true your hypothesis is. Second, use a t-test unless someone literally hands you σ , which they won’t. Third, significant and important are not the same word.



You started at zero and finished with z-tests, t-tests, p-values, confidence intervals, and the ability to call out bad science at dinner parties. Not bad for one article.

Statistics

Data Science

Python

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Hypothesis Testing



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MSc in Applied Statistics & Data Science. I write about machine learning, deep learning, and statistics with Python, making complex ideas ridiculously clear.
